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MANIPAL INSTITUTE
OF TECHNOLOGY

(A constituent unit of MAHE, Manipal)

CLOUD COMPUTING and DEVOPS (CSE 3130)
LAB MANUAL
5th Semester, B. TECH

School of Computer Engineering

SCHOOL OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that Ms./Mr.

Reg. No.: Section: Roll No.:

has satisfactorily completed the lab exercises prescribed for Cloud Computing & DevOps of Fifth Semester B.Tech, Degree at MIT, Manipal, in the academic year 2026-2027.

Date:

Signature of the Faculty

Signature of Dean

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Table of Contents

COURSE OBJECTIVES	3
COURSE OUTCOMES (COs)	4
EVALUATION PLAN	4
INSTRUCTIONS TO THE STUDENTS	5
Pre-Lab Session Instructions:	5
In-Lab Session Instructions:	5
General Instructions for the exercises in Lab:	5
The Students Should Not	6
LAB EXPERIMENTS	7
LAB 01: Configure Nginx Webserver on local Linux Machine to serve static and web pages via PHP and JSP	7
LAB 02: Provisioning an isolated virtualization environment using Oracle Virtual Box & deploy 3-tier LEMP stack solution	14
REFERENCES:	19

COURSE OBJECTIVES

At the end of this course, the student should be able to:	No. of Contact	Marks	Program Outcomes	Learning Outcomes	BL (Recomm
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		Hours		(POs)	(LOs)	ended)
CL O1	To develop practical skills in version control strategies, automated continuous integration/continuous deployment (CI/CD) pipelines, and Infrastructure as Code (IaC) integration for DevOps workflows.					
CL O2	To build proficiency in deploying, configuring, and managing containerized applications through container orchestration techniques to ensure system scalability and fault tolerance.					
CL O3	To examine system performance monitoring, log analysis techniques, and the integration of cloud-native observability tools for effective operational incident response.					
	Total					

COURSE OUTCOMES (COs)

EVALUATION PLAN

Continuous Evaluation	60%	
1. Lab Examination	40%	

INSTRUCTIONS TO THE STUDENTS

Pre-Lab Session Instructions:

1. Students should carry the Lab Manual Book and the required stationery to every lab session
2. Be in time and follow the institution dress code
3. Must sign in the log register provided
4. Make sure to occupy the allotted seat and answer the attendance
5. Adhere to the rules and maintain the decorum

In-Lab Session Instructions:

- Follow the instructions on the allotted exercises
- Show the program and results to the instructors on completion of experiments
- On receiving approval from the instructor, copy the program and results in the lab record
- Prescribed textbooks and class notes can be kept ready for reference if required

General Instructions for the exercises in Lab:

- Implement the given exercise individually and not in a group.
- The programs should meet the following criteria:
 - Programs should be interactive with appropriate prompt messages, error messages if any, and descriptive messages for outputs.
 - Programs should perform input validation (Data type, range error, etc.) and give appropriate error messages and suggest corrective actions.
 - Comments should be used to give the statement of the problem and every function should indicate the purpose of the function, inputs and outputs.
 - Statements within the program should be properly indented.
 - Use meaningful names for variables and functions.
 - Make use of constants and type definitions wherever needed.
- Plagiarism (copying from others) is strictly prohibited and would invite severe

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penalty in evaluation.

- In case a student misses a lab class, he/she must ensure that the experiment is completed during the repetition class in case of genuine reason (medical certificate approved by HOD) with the permission of the faculty concerned
- Questions for lab tests and examination are not necessarily limited to the questions in the manual, but may involve some variations and/or combinations of the questions.
- A sample note preparation is given as a model for observation.

The Students Should Not

Bring mobile phones or any other electronic gadgets to the lab.

Go out of the lab without permission.

LAB EXPERIMENTS

LAB 01: Configure Nginx Webserver on local Linux Machine to serve static and web pages via PHP and JSP

Objectives:

1. To be familiar with Linux commands
2. To learn about web servers Nginx , PHP-FPM , Java Container Tomcat
3. To understand how to configure a web server to serve static and web pages via different programming languages
4. To learn to check the server logs

Conceptual Background:

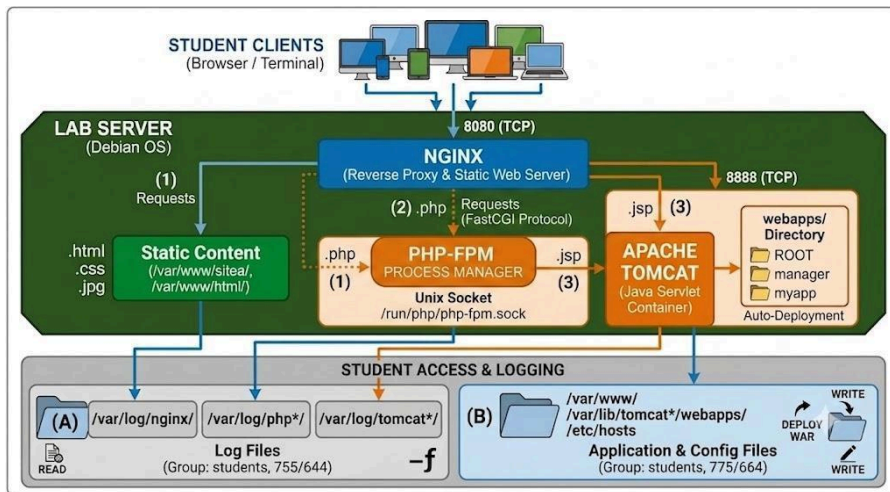


Figure 1: Lab1 setup

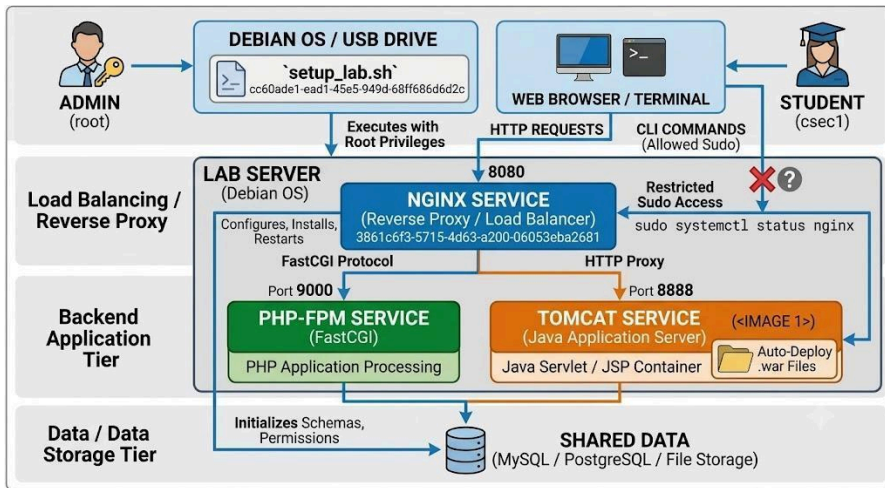


Figure 2: Typical 3-tier architecture

Nginx: Event-Driven Reverse Proxy & Web Server

Unlike traditional web servers (like Apache httpd) that spawn a dedicated thread or process for every incoming connection (a thread-per-connection model), Nginx utilizes an asynchronous, event-driven, non-blocking architecture.

Key Architectural Concepts:

- **Master-Worker Architecture:** Nginx runs one Master Process (which reads configurations and maintains worker processes) and multiple Worker Processes (which handle actual network connections).
- **Event Loop:** A single worker process uses OS-level I/O multiplexing mechanisms (such as `epoll` in Linux) to handle thousands of concurrent HTTP connections on a single thread with minimal RAM and CPU overhead.
- **Primary Roles:**
 - **Static File Server:** Serves HTML, CSS, JavaScript, and images directly from disk with extreme speed.
 - **Reverse Proxy:** Accepts client requests on port 8080 (or 80) and forwards them to backend application servers (like PHP-FPM or Tomcat).
 - **Load Balancer:** Distributes incoming traffic across multiple backend application instances.

PHP Application Processing: FastCGI & PHP-FPM

Nginx cannot process dynamic code (like `.php` scripts) natively. It only acts as a delivery mechanism. To execute PHP code, Nginx relies on an external process manager using the FastCGI protocol.

Key Architectural Concepts

1. CGI vs. FastCGI:

- CGI (Common Gateway Interface): Spawns a new OS process for every single HTTP request, leading to massive memory overhead.
 - FastCGI: Keeps long-running worker processes alive in memory to execute incoming scripts sequentially, avoiding process creation overhead.
2. PHP-FPM (FastCGI Process Manager):
- PHP-FPM is an advanced implementation of FastCGI for PHP.
 - It maintains a dynamic pool of worker processes waiting to accept requests.
3. How Nginx Interacts with PHP-FPM:
- Client requests `http://localhost:8080/index.php`.
 - Nginx checks the location block matching `.php`.
 - Nginx translates the HTTP request into FastCGI binary format and sends it over a Unix Domain Socket or TCP/IP Socket (typically `127.0.0.1:9000`) to PHP-FPM.
 - PHP-FPM executes the script, captures output, and returns HTML back to Nginx, which sends it to the browser.

Apache Tomcat: Java Servlet & JSP Container

Apache Tomcat is an open-source Java Servlet Container (and Web Server) that implements Jakarta Servlet, Jakarta Expression Language, and WebSocket technologies. Unlike Nginx or standard Apache HTTP, Tomcat is specifically designed to execute compiled Java code packaged inside `.war` (Web Application Archive) files.

Key Architectural Concepts

- Servlet Container (Catalina): Catalina is the core engine of Tomcat that handles request lifecycle management, routing incoming requests to the appropriate Java Servlet class, instantiating Java objects, and managing sessions.
- JSP Engine (Jasper): Converts JavaServer Pages (`.jsp`) into standard Java Servlets on-the-fly, compiles them into bytecodes (`.class`), and executes them.
- Deployment (`.war` Files): A `.war` file is a compressed ZIP archive containing compiled Java bytecode, web configuration descriptors (`web.xml`), and static assets. Tomcat automatically uncompresses and deploys any `.war` file placed into its `webapps/` folder.
- Architecture Role: Tomcat usually listens on a dedicated internal port (like 8888) and handles business logic/database operations. Nginx sits in front of Tomcat to handle TLS termination, static asset caching, and security filtering.

Solved Examples:

General Instructions:

- Perform all steps logged into your assigned student user account (e.g., `csec1`). Do not log in as root.
- When managing services or network interfaces, prefix commands with `sudo`. You will not be prompted for a password for authorized lab commands.

- You may use nano or vi in the terminal or launch VSCode to edit configuration files.
- Nginx (8080) and Tomcat (8888) is already installed on your machine and running at specific ports mentioned.

1) Configure Nginx webserver to serve static web page (Name-Based Virtual Hosting)

Name-based virtual hosting allows a single Nginx web server instance to host multiple domain names (e.g., sitea.local) on a single IP address.

Step 1.1: Map Domain Names Locally

Since local development domain names are not registered on public DNS servers, map sitea.local to your local loopback address (127.0.0.1).

- Open /etc/hosts with administrative rights:
`nano /etc/hosts`
- Add the following line at the end of the file:
`127.0.0.1 sitea.local`

Step 1.2: Create the Web Document Root

- Create a dedicated folder for sitea.local and add an HTML entry point.

Create the web root directory:

```
mkdir -p /var/www/sitea
```

- Create an index.html file inside the new folder:

```
nano /var/www/sitea/index.html
```

Paste the following HTML sample:

```
<!DOCTYPE html>
<html>
<head>
    <title>Welcome to Site A</title>
</head>
<body>
    <h1>Success! Name-Based Virtual Host (Site A) is
working!</h1>
</body>
</html>
```

Step 1.3: Create the Nginx Server Block Configuration

- Define how Nginx routes web traffic for sitea.local.

Create a new server block file in sites-available:

```
nano /etc/nginx/sites-available/sitea.conf
```

Add the following configuration block:

```
server {
    listen 8080;
    server_name sitea.local;

    root /var/www/sitea;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
```

Step 1.4: Enable the Configuration and Reload Nginx

- Link the configuration file to the active sites-enabled directory and apply changes.

Create a symbolic link to activate the site:

```
ln -s /etc/nginx/sites-available/sitea.conf
/etc/nginx/sites-enabled/
```

- Verify that your configuration syntax contains no errors:

```
sudo nginx -t
```

- Reload Nginx to apply the new Virtual Host:

```
sudo systemctl reload nginx
```

Step 1.5: Verification

- Open web browser and navigate to:
`http://sitea.local:8080`
- Alternatively, test the HTTP response via terminal:
`curl http://sitea.local:8080`
(Expected output: The HTML content created in Step 1.2).

Step 1.6: View error and access logs

- view errors

```
tail -f /var/log/nginx/error.log
```

- view incoming requests

```
tail -f /var/log/nginx/access.log
```

2) Configure Nginx webserver to serve Dynamic Pages via PHP(FastCGI/PHP-FPM)

Nginx processes dynamic scripts by passing requests to an external application processing pool (PHP-FPM) via the FastCGI protocol.

Step 2.1: Create the PHP Web Document Root

- Create a directory for your PHP web files:

```
mkdir -p /var/www/php_site
```

- Create an index.php test file:

```
nano /var/www/php_site/index.php
```

- Add code to display detailed PHP runtime environment information:

```
<?php
    echo "<h1>Nginx + PHP-FPM FastCGI Execution Successful</h1>";
    phpinfo();
?>
```

Step 2.2: Identify the Installed PHP-FPM Socket

- Nginx communicates with PHP-FPM through a Unix domain socket. Identify the active PHP version socket on your machine:

```
ls /run/php/php*-fpm.sock
```

(Example output path: /run/php/php8.2-fpm.sock). Note your exact version number.

Step 2.3: Configure Nginx for FastCGI Processing

- Create a new configuration file for the dynamic site:

```
nano /etc/nginx/sites-available/php_site.conf
```

- Add the configuration block, updating the socket path to match Step 2.2:

```
server {
```

```

listen 8080;
server_name phpsite.local;

root /var/www/php_site;
index index.php index.html;

location / {
    try_files $uri $uri/ =404;
}

# Pass PHP scripts to FastCGI server via PHP-FPM Unix socket
location ~ \.php$ {
    include snippets/fastcgi-php.conf;
    fastcgi_pass unix:/run/php/php8.2-fpm.sock;
}
}

```

- Map phpsite.local in /etc/hosts:

```
nano /etc/hosts
```

Add following at the end of the file

```
127.0.0.1 phpsite.local
```

Step 2.4: Enable Site and Test Configuration

- Enable the PHP site:

```
ln -s /etc/nginx/sites-available/php_site.conf
/etc/nginx/sites-enabled/
```

- Check configuration syntax:

```
sudo nginx -t
```

- Reload Nginx:

```
sudo systemctl reload nginx
```

Step 2.5: Verification

- Open your browser and navigate to: <http://phpsite.local:8080>

Confirm that the page renders the dynamic PHP Version Information dashboard.

- Alternatively, test the HTTP response via terminal:

```
curl http://phpsite.local:8080
```

(Expected output: It should render the contents mentioned in Step 2.1 with php server configuration list).

Step 2.6: View error and access logs

- view errors

```
tail -f /var/log/nginx/error.log
```

- view incoming requests

```
tail -f /var/log/nginx/access.log
```

Exercises:

Record your findings and understandings in the observation book

- 1) Configure Nginx webserver to server static web page(IP-Based Virtual Hosting)

Notes:

- You can use secondary IP Address: assign 127.0.0.2 (or 127.0.0.3) as your secondary IP.
- You have sudo permissions for ip addr add, you can run following
sudo ip addr add 127.0.0.2/8 dev lo

- 2) Configure Nginx webserver to server dynamic Pages via JSP(OpenJDK + Tomcat Reverse Proxy)

Notes:

- Tomcat listens on port 8888. Configure Nginx to proxy requests on port 8080 to Tomcat's port 8888
- If you deploy a sample.war file or a raw JSP folder directly, it goes into /var/lib/tomcat10/webapps/
- If JSP app fails, you can check logs at /var/log/tomcat10/catalina.out.

Remarks:

After completion of Exercises, run the following (mandatory)

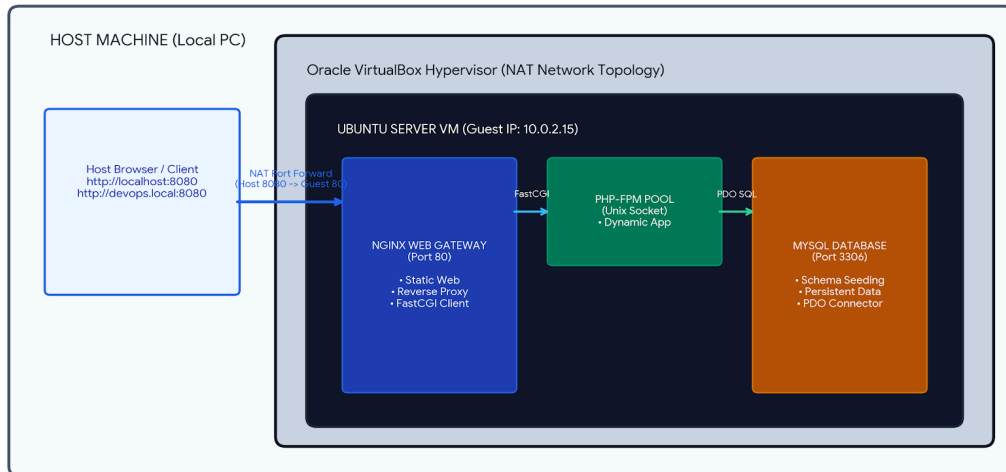
```
sudo lab-reset
```

LAB 02: Provisioning an isolated virtualization environment using Oracle Virtual Box & deploy 3-tier LEMP stack solution

Objectives:

- To gain hands-on experience with hypervisor-based virtualization using Oracle Virtual Box.
- To configure guest-to-host virtual network topologies using NAT and Port Forwarding.
- To deploy and manage a multi-tier LEMP stack (Linux, Nginx, MySQL, PHP-FPM) inside an isolated guest environment.
- To examine Layer 7 name-based ingress routing, process failure modes, and system resource monitoring under load.

Conceptual Background:



1. Hardware Virtualization with Type-2 Hypervisor:

While Lab 01 configured web daemons directly on local host operating systems, modern cloud infrastructure relies on hypervisor. A Type-2 Hypervisor (such as Oracle Virtual Box) runs on top of a host operating system and virtualizes hardware resources (CPUs, Memory, Storage, and Virtual Network Adapters) to run fully isolated guest operating systems (For example Ubuntu Desktop or Server).

2. Network Virtualization (NAT and Port Forwarding):

By default, Virtual Box isolates virtual machines behind a Network Address Translation (NAT) virtual router. The virtual machine receives an isolated internal IP address (e.g., 10.0.2.15) that cannot be accessed directly from the host network or external client browsers. To enable access to guest web applications, NAT Port Forwarding establishes a Layer 4 socket mapping rule at the hypervisor level. Requests sent to a designated port on the physical host (e.g., 127.0.0.1:8080) are intercepted by the Virtual Box process and transparently translated to port 80 inside the virtual machine instance.

3. 3-Tier LEMP Stack Architecture:

A production LEMP stack partitions dynamic web hosting into 3 layers:

- **Presentation/Ingress Layer (Nginx):** An asynchronous, event-driven web gateway listening on Port 80. It serves static assets directly and routes dynamic execution

requests via the FastCGI protocol.

- Application Layer (PHP-FPM): The FastCGI Process Manager runs an isolated worker pool communicating with Nginx over a local Unix domain socket (/var/run/php/php8.3-fpm.sock), avoiding process-spawning overhead.
- Data Persistence Layer (MySQL Server): A relational database engine listening on Port 3306 that manages relational schemas, user privileges, and persistent records via dynamic PHP PDO drivers.

4. Cloud Ingress and Name-based Virtual hosting:

In modern cloud environments (such as Kubernetes Ingress Controllers or AWS Application Load Balancers), a single virtual gateway instance routes incoming traffic to multiple backend microservices based on the **HTTP Host: header**. Nginx matches incoming host request strings (e.g., devops.local vs. analytics.local) to distinct server { ... } configuration blocks, providing multi-tenant application routing on a single public port.

Solved Examples:

Step 1: Virtual Machine Provisioning in Oracle Virtual Box

- Launch Oracle Virtual Box and click New.
- Configure
 - VM Name: DevOps_Lab2_LEMP
 - ISO Image: choose the ISO image downloaded from Lab shared server
 - OS: Linux
 - OS Distribution: Ubuntu
 - OS Version: Ubuntu (64-bit).
 - Base Memory: 4000 MB RAM (4 GB)
 - Number of CPUs: 12 CPUs
 - Disk Size: 25 GB

Step 2: Hypervisor NAT Network & Port Forwarding Setup

- Navigate to VM Settings > Network > Adapter 1.
- Select Attached to: NAT
- Expand Advanced, click Port Forwarding, and insert the HTTP mapping rule:
 - Name: HTTP
 - Protocol: TCP
 - Host Port: 8080
 - Guest Port: 80

Step 3: Guest OS Package Deployment

- Start the VM, complete the standard Ubuntu OS installation wizard, create user **student/ password** and log into the guest shell.

Note: if you encounter any error while starting the VM, close Oracle Virtual Box , then execute following in **Host OS** terminal

- `sudo apt update && sudo apt install -y linux-headers-amd64`
`linux-headers-$(uname -r) build-essential dkms`
- `sudo /sbin/vboxconfig`

- `echo "blacklist kvm_intel" | sudo tee -a /etc/modprobe.d/blacklist.conf`
- `echo "blacklist kvm" | sudo tee -a /etc/modprobe.d/blacklist.conf`
- Start Ubuntu (Guest OS) VM
- Inside Guest OS
 - Update repository indexes and install Nginx, PHP-FPM, and MySQL server:
`sudo apt update && sudo apt install nginx php-fpm php-mysql mysql-server -y`
 - Verify daemon execution status:
`systemctl status nginx`
`systemctl status php*-fpm`
`systemctl status mysql`

Step 4: Nginx FastCGI Server Block Configuration

- Edit the default server block file:
`sudo nano /etc/nginx/sites-available/default`
- Replace content with the FastCGI processing block:

```
server {
    listen 80 default_server;
    listen [::]:80 default_server;
    root /var/www/html;
    index index.php index.html index.htm;
    server_name _;
    location / {
        try_files $uri $uri/ =404;
    }
    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/var/run/php/php8.3-fpm.sock; # Adjust socket version if required
    }
    location ~ /\.ht {
        deny all;
    }
}
```
- Validate syntax and reload Nginx:
`sudo nginx -t && sudo systemctl reload nginx`

Exercises:

Exercise #1: Hypervisor Network Isolation & Gateway Address Mapping

- Create /var/www/html/server_info.php inside the VM:

```
<?php
    echo "<h3>Server Infrastructure Details</h3>";
    echo "VM Guest IP Address: " . $_SERVER['SERVER_ADDR'] . "<br>";
    echo "Client Remote Access IP: " . $_SERVER['REMOTE_ADDR'] . "<br>";
    echo "Gateway Server Port: " . $_SERVER['SERVER_PORT'];
?>
```
- Access http://localhost:8080/server_info.php from your **Host machine browser** and record findings.
 - VM Guest Internal IP (`$_SERVER['SERVER_ADDR']`):

- Client Remote IP (\$_SERVER['REMOTE_ADDR']):
- Explain how VirtualBox NAT Port Forwarding routes Host Port 8080 requests to Guest Port 80:

Exercise #2: Analyse Server/Process failure

- Stop PHP-FPM daemon (sudo systemctl stop php*-fpm) and refresh http://localhost:8080/server_info.php in host browser.
- Restart PHP-FPM (sudo systemctl start php*-fpm) and stop Nginx (sudo systemctl stop nginx).
- Record following findings
 - Error displayed & Error thrown in error.log when PHP-FPM is stopped
 - Error displayed when Nginx service is stopped

Exercise #3: System Resource Utilization & Workload Monitoring

- Open a second terminal inside the VM and launch process monitor: htop (or top).
- Execute 500 requests from host machine command line:

```
for i in {1..500}; do curl -s http://localhost:8080/server_info.php > /dev/null; done
```
- Record following findings
 - Baseline CPU / Memory Usage: % / MB
 - Peak Workload CPU / Memory Usage: % / MB
 - Which process (nginx worker vs php-fpm worker) registered higher CPU utilization spike?

Additional Exercises(optional):

Exercise #4: 3-Tier Database Connectivity & Data Persistence

- Access MySQL CLI inside VM (sudo mysql) and seed schema:

-- 1. Create the lab database

```
CREATE DATABASE devops_lab;
```

-- 2. Create a dedicated database user with a secure password

```
CREATE USER 'lab_user'@'localhost' IDENTIFIED BY 'LabPassword123!';
```

-- 3. Grant full permissions on devops_lab database to lab_user

```
GRANT ALL PRIVILEGES ON devops_lab.* TO 'lab_user'@'localhost';
```

```
FLUSH PRIVILEGES;
```

-- 4. Switch to the newly created database

```
USE devops_lab;
```

-- 5. Create the table schema

```
CREATE TABLE student_records (
```

```
  id INT AUTO_INCREMENT PRIMARY KEY,
```

```
  student_name VARCHAR(50) NOT NULL,
```

```
  roll_no VARCHAR(20) NOT NULL
```

```
);
```

-- 6. "SEED" the schema (Insert sample initial records)

```
INSERT INTO student_records (student_name, roll_no) VALUES ('Arun', '1001'), ('Joesph', '1002');
EXIT;
```

- Create /var/www/html/db_test.php using PDO connection logic

```
<?php
    $dsn = "mysql:host=127.0.0.1;dbname=devops_lab;charset=utf8mb4";
    try {
        $pdo = new PDO($dsn, 'lab_user', 'LabPassword123!');
        echo "<h3 style='color:green;'>MySQL Database Connection Successful!</h3>";
        $stmt = $pdo->query('SELECT id, student_name, roll_no FROM student_records');
        echo "<table border='1'><tr><th>ID</th><th>Name</th><th>Roll No</th></tr>";
        while ($row = $stmt->fetch()) {
            echo
            "<tr><td>{$row['id']}</td><td>{$row['student_name']}</td><td>{$row['roll_no']}</td></tr>";
        }
        echo "</table>";
    } catch (PDOException $e) {
        echo "<h3 style='color:red;'>Database Connection Failed: ". $e->getMessage() . "</h3>";
    }
?>
```

- Verify at http://localhost:8080/db_test.php
- Record following findings
 - Database Query Table Rendering Status: [] SUCCESS [] FAILURE
 - Resulting PDOException output when MySQL service is stopped (sudo systemctl stop mysql):

Exercise #5: Cloud Ingress Routing — Multi-Site Name-Based Virtual Hosting

- Create web roots /var/www/devops_app and /var/www/analytics_app with corresponding index files.
- Configure site blocks for devops.local and analytics.local under /etc/nginx/sites-available/
- Create Symbolic links both site files to /etc/nginx/sites-enabled/, remove default site, and reload Nginx.
- Add loopback entries (127.0.0.1 devops.local analytics.local) into Host Machine hosts file.
- Verify <http://devops.local:8080> and <http://analytics.local:8080> in host browser.
- Record following findings
 - Specific HTTP Request Header used by Nginx to differentiate virtual site hosts:

Remarks

- Power off the Virtual Machine: Inside the Ubuntu VM terminal, run:
`sudo poweroff`
- Delete the VM Instance and Disk Files:
 - Open Oracle VM VirtualBox Manager on your host machine.
 - Right-click on the DevOps_Lab2_LEMP virtual machine in the left panel.
 - Select Remove
 - Click Delete all files (this permanently removes the virtual hard disk .vdi file and

associated VM settings from the lab machine).

REFERENCES:

1	